📘 TITLE: Advanced Python Programming for Competitive and GATE-Level Understanding

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1. Introduction to Python

Python is a high-level, interpreted, dynamically typed programming language known for its readability and simplicity. It supports multiple paradigms including procedural, object-oriented, and functional programming.

Key characteristics:

- Interpreted execution
- Dynamic typing
- Automatic memory management
- Extensive standard library

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2. Python Execution Model

Python code is executed in the following steps:

1. Source Code (.py)
2. Compilation → Bytecode (.pyc)
3. Execution via Python Virtual Machine (PVM)

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3. Data Types and Mutability

3.1 Immutable Types

- int
- float
- str
- tuple

3.2 Mutable Types

- list
- dict
- set

Example:

```
a = [1, 2, 3]
b = a
```

```
b.append(4)
print(a)
```

Output:

[1, 2, 3, 4]

Explanation: Lists are mutable; both variables reference same object.

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4. Variable Binding and Memory Model

Python uses **reference binding**, not traditional variable storage.

```
x = 10
y = x
x = 20
print(y)
```

Output:

10

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5. Deep vs Shallow Copy

Shallow Copy

- Copies reference

Deep Copy

- Copies full object

```
import copy

a = [[1, 2], [3, 4]]
b = copy.copy(a)
c = copy.deepcopy(a)
```

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6. Functions and Scope

LEGB Rule:

- Local
- Enclosing
- Global
- Built-in

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7. Recursion

A function calling itself.

Example:

```
def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n-1)
```

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8. Time Complexity (Important for GATE)

Operation	Time Complexity
List append	O(1)
List search	O(n)
Dict access	O(1) average
Sorting	O(n log n)

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9. Python Collections

List

Ordered, mutable

Tuple

Ordered, immutable

Set

Unordered, unique elements

Dictionary

Key-value pairs

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10. List Comprehension

```
squares = [x*x for x in range(10)]
```

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11. Generators

Generators produce values lazily.

```
def gen():
    for i in range(5):
        yield i
```

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12. Exception Handling

```
try:
    x = 10 / 0
except ZeroDivisionError:
    print("Error")
```

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13. Object-Oriented Programming

Concepts:

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

Example:

```
class A:
    def show(self):
        print("Class A")

class B(A):
    pass
```

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14. Method Resolution Order (MRO)

Defines order in which base classes are searched.

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15. Iterators vs Generators

Feature	Iterator	Generator
Memory	High	Low
Syntax	Class	yield

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16. Lambda Functions

```
f = lambda x: x + 1
```

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17. Functional Programming

- map()
- filter()
- reduce()

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18. File Handling

```
with open("file.txt", "r") as f:
    data = f.read()
```

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19. Decorators

Functions that modify other functions.

```
def decorator(func):
    def wrapper():
        print("Before")
        func()
        print("After")
    return wrapper
```

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20. Multithreading vs Multiprocessing

Feature	Threading	Multiprocessing
GIL	Present	Not limiting
Speed	Limited	Faster CPU tasks

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21. Python Memory Management

- Reference counting
- Garbage collection

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22. Advanced Concepts

- Closures
- Context Managers
- Metaclasses

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23. Common GATE-Level Traps

- Mutable default arguments
- Late binding in closures
- Shallow copy issues

📘 TITLE: Advanced Python Programming — Theory, Internals, and Computational Concepts

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1. Python as a Programming Paradigm

Python is a **multi-paradigm programming language** that supports procedural, object-oriented, and functional programming styles. Unlike purely compiled languages, Python emphasizes readability, abstraction, and rapid development.

At its core, Python is dynamically typed, meaning type checking occurs at runtime rather than compile time. This flexibility allows for faster development but introduces potential runtime errors if not handled carefully.

Python follows the philosophy of:

- Code readability
- Simplicity over complexity
- Explicit over implicit

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2. Internal Working of Python (Execution Model)

Python programs do not execute directly as machine code. Instead, they follow a multi-stage execution process:

1. Source code (.py file) is written by the developer
2. Python compiler converts it into bytecode
3. Bytecode is executed by the Python Virtual Machine (PVM)

The bytecode is platform-independent, which contributes to Python's portability.

Important Insight:

The PVM acts as an interpreter for bytecode, making Python slower than compiled languages like C but more flexible.

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3. Python Memory Model and Object System

Python uses a **private heap** to store all objects and data structures. The Python memory manager controls allocation and deallocation.

Key Concepts:

- Every variable is a reference to an object
- Objects contain:
    - identity (memory address)
    - type
    - value

Example Insight:

```
x = [1, 2]
y = x
```

Both x and y point to the same memory location.

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4. Mutability and Its Implications

Mutability determines whether an object's value can be changed after creation.

Immutable Objects:

- int, float, str, tuple

Mutable Objects:

- list, dict, set

Critical GATE Concept:

Mutable objects can lead to unintended side effects when passed across functions or assigned to multiple variables.

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5. Variable Binding and Scope Resolution

Python does not use traditional variable storage. Instead, it uses **name binding**.

LEGB Rule:

- Local
- Enclosing
- Global
- Built-in

Deep Insight:

Closures retain access to variables from enclosing scopes even after the outer function has finished execution.

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6. Functions as First-Class Objects

In Python:

- functions can be assigned to variables
- passed as arguments
- returned from other functions

Example:

```
def outer():
    def inner():
        return "Hello"
    return inner
```

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7. Recursion and Stack Behavior

Recursion uses the call stack to maintain function states.

Important Points:

- Each recursive call consumes memory
- Python has a recursion depth limit (~1000)

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8. Python Data Structures — Deep Analysis

Lists

- Dynamic arrays
- Amortized O(1) append

Dictionaries

- Hash tables
- Average O(1) lookup

Sets

- Unique elements
- Fast membership testing

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9. Hashing and Dictionary Internals

Python dictionaries use hashing to map keys to values.

Key Properties:

- Keys must be immutable
- Hash collisions handled internally

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10. Time Complexity Analysis

Operation	Complexity
List append	O(1)
List insert	O(n)
Dict lookup	O(1) avg
Sorting	O(n log n)

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11. Iterators and Generators

Iterators:

- Implement **iter**() and **next**()

Generators:

- Use yield
- Memory efficient

Insight:

Generators are preferred for large datasets because they avoid loading everything into memory.

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12. Exception Handling Mechanism

Python uses structured exception handling:

```
try:
    pass
except:
    pass
finally:
    pass
```

Deep Concept:

Exception handling improves robustness but can impact performance if overused.

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13. Object-Oriented Programming in Python

Core Concepts:

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

Method Resolution Order (MRO)

Python uses C3 linearization to resolve method lookup.

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14. Functional Programming Concepts

Python supports:

- map()
- filter()
- reduce()

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15. Closures and Late Binding

Closures capture variables, but Python uses **late binding**, meaning variables are evaluated when used, not when defined.

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16. Decorators (Advanced Concept)

Decorators modify function behavior without changing its code.

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17. Concurrency in Python

Multithreading:

- Limited by Global Interpreter Lock (GIL)

Multiprocessing:

- True parallelism

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18. Global Interpreter Lock (GIL)

GIL ensures only one thread executes Python bytecode at a time.

Implication:

- CPU-bound tasks → multiprocessing
- I/O-bound tasks → threading

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19. Memory Management

Python uses:

- Reference counting
- Garbage collection

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20. Advanced Topics

- Metaclasses
- Context Managers
- Async programming

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21. Common GATE-Level Pitfalls

- Mutable default arguments
- Shallow vs deep copy confusion
- Late binding in loops

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22. Real-World Optimization Case

A system replaced list processing with generators and reduced memory usage significantly.

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23. Extended Theory Section (Dense Content for RAG)

Python's flexibility comes from its dynamic typing and runtime behavior, but this also introduces complexity in understanding memory and execution flow. Advanced users must deeply understand reference semantics, object mutability, and performance trade-offs.

Understanding internal mechanisms such as hashing, garbage collection, and execution models is crucial for solving complex computational problems efficiently.